

Design and Implementation of a Dual-Evaporator Transcritical High Temperature Heat Pump

Research Project

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1 Context

According to the IEA report from March 2025, the global CO₂ emissions have increased by 0.8% in 2024, reaching an all-time high of 37.8 Gt CO₂ a year. In addition, the report highlights that the energy demand grew by 2.2% in 2024. This increase in CO₂ emissions and energy demand is totally inconsistent with the goals of the Paris Agreement, which aims to abate the CO₂ emissions by 43% by 2030 and to reach net-zero emissions by 2050.

In [2], we can read that the waste heat potential in the EU-27 and UK reached 221.32 TWh/year in 2021. In comparison, the demand for less than 140°C heat in Germany was 612 PJ, or 170 TWh in 2012 [1]. These numbers justify the interest from companies/researchers to design and study machines capable of recovering this waste energy.

In this work, we will focus on the use of high temperature heat pumps (HTHP's) for recovering waste heat from industrial processes. In [1], the authors provide a list of manufacturers and universities that have already been working on the design of several machines for at least one decade. The already existing machines cover a large spectrum of applications, going from small machines of a few thousands watts to big ones of more than one megawatt. However, new design ideas can still be studied in order to improve the current recovery systems.

The main purpose of this work is to design the thermodynamical cycle of a trans-critical dual-evaporator HTHP, build it and test its performances in laboratory conditions. No similar applications were found in the literature. Beside, the machine build in the frame of this master thesis should be used in the future for an academic purpose.

2 Literature review

To be completed by FG (2 pages).

3 Research question

To be completed by HW (1 page).

4 Project description

In order to answer to the previous research question, we distinguish six steps in this master thesis. These steps can be observed in the Figure ...

Preliminary cycle design

In this first part, we will numerically modeled the thermodynamical design. Each student will work simultaneously on different blocks, determined in advance. The analysis of the cycle will include an energy and exergy analysis to ensure optimal performances. The numerical model will be written in Python and the module Coolprop will be used to access numerical thermodynamics tables.

Final cycle design

Assembling all the different blocks will be the purpose of the second part. Beside connecting the blocks, an optimization process will be developped to determine the optimal parameters in nominal conditions. In addition, more physical aspects will be added to the model such as head losses computations and heat transfer modeling.

Control

One main advantage of this machine will be its capacity of simulating different conditions thanks to the dual evaporator. A control logic must therefore be implemented. This part is critical because we'll have to think about the nature and the location of the sensors. A student in industrial engineering will bring his expertise and will start to collaborate with us.

Machine design

In this part, the other student will be in charge of the design of the machine. It will contain material/components selection, heat exchangers and insulation design and the elaboration of precise technical draw. That will be our task to correct the final cycle design (head losses, heat exchange, ...). We will also collaborate with Frank, the technician that will be responsible of the buiding of the phase. We surely think that its expertise will be needed for many design uncertainties.

Building

The building phase will be ensured by Frank and other technicians of the CREDEM.

Testing

Finally, the last part will be dedicated to the evaluation of the machine. Several tests will take place to ensure the good operation of the machine and to collect data in order to compare them to theoretical values. We don't think it will possible to have a correction phase in case of underperforming.

5 Share of workload and responsibilities

To be completed by FG (1 page).

6 Roadmap and planning

To be completed by HW (2 pages).

References

- [1] Cordin Arpagaus, Frédéric Bless, Michael Uhlmann, Jürg Schiffmann, and Stefan S Bertsch. High temperature heat pumps: Market overview, state of the art, research status, refrigerants, and application potentials. *Energy*, 152:985–1010, 2018.
- [2] George Kosmadakis. Industrial waste heat potential and heat exploitation solutions. *Applied Thermal Engineering*, 246:122957, 2024.